

SE200 Electrical Submersible Pump (ESP)

Version 1.0

Operations Manual

SOFTWARE VERSION 1.11_4

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Table of Contents

Keypad (HMI)	3
Power and Motor Terminations	4
Power Input Connections	4
Motor Wiring	4
Sizing Requirement	Error! Bookmark not defined.
Parameter Configuration	Error! Bookmark not defined.
Control Wiring.....	5
Initial Power Up	12
Main Screen	17
Status Page.....	19
Motor Setup.....	20
I/O & DH Tool Status.....	21
Config 1 (VFD S/D).....	22
Config 2 (DH Tool S/D)	23
Shutdown Test	24
Trending #1 (VFD Info).....	25
Trending #2 (DH Tool Info).....	26
Trending Scales	27
Alarm History	27
Login Pop-up	29
Troubleshooting.....	30

Keypad (HMI)

The SE200 HMI main screen provides the primary controls and status indicators used to operate and monitor the VFD. An example of the main screen is shown in the figure below.

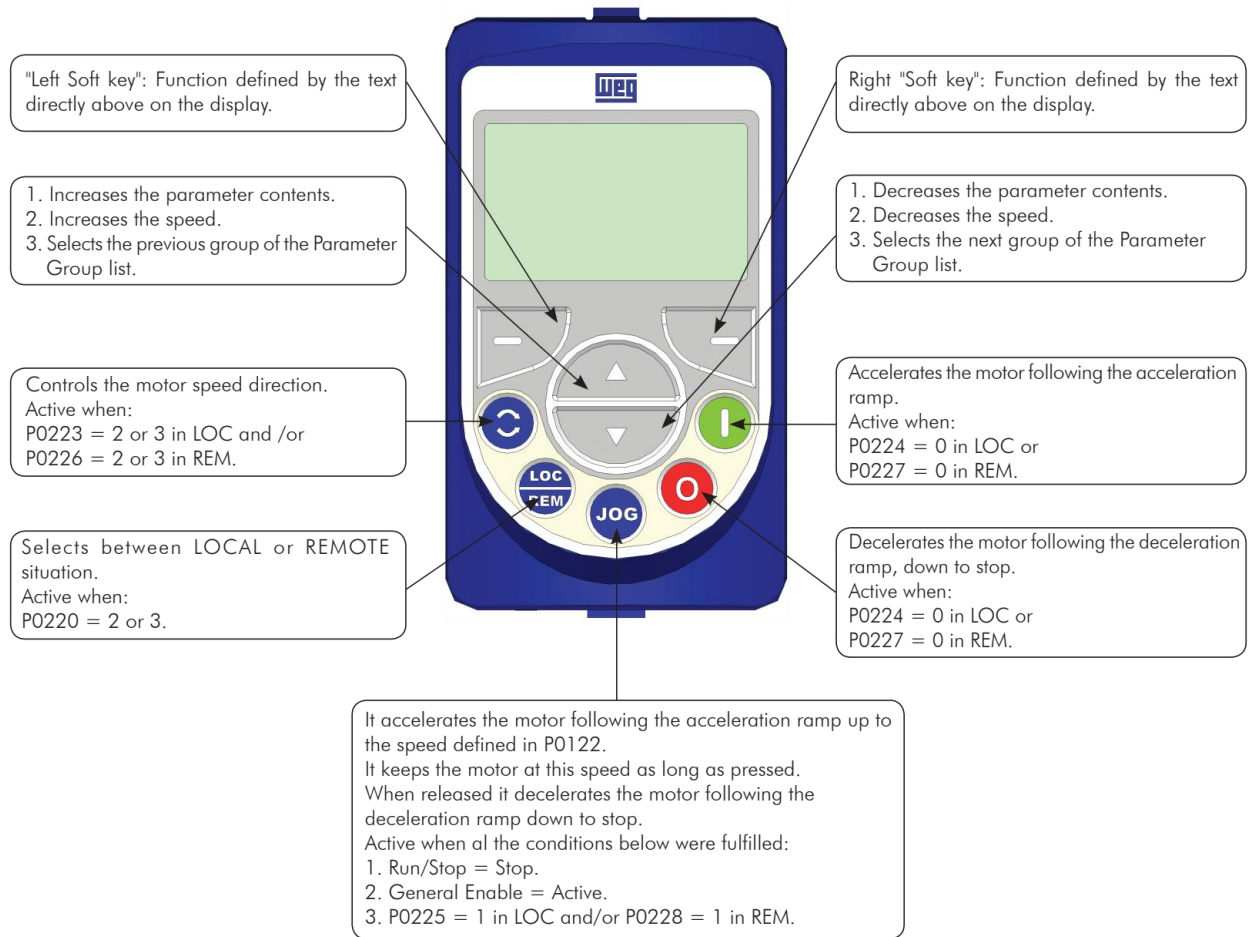


Figure 1 Keypad

Power and Motor Terminations

The SE200 ESP VFD can only be used in three phase applications.

1. Power Input Connections

- For three-phase power connect A/L1 (**Red**), B/L2 terminals (**Black**) and C/L3 (**Blue**).
- These terminals supply power to the DC bus of the CFW11.

2. Motor Wiring

- Terminate the motor leads at the U, V, and W terminals, located at the bottom right of the VFD.

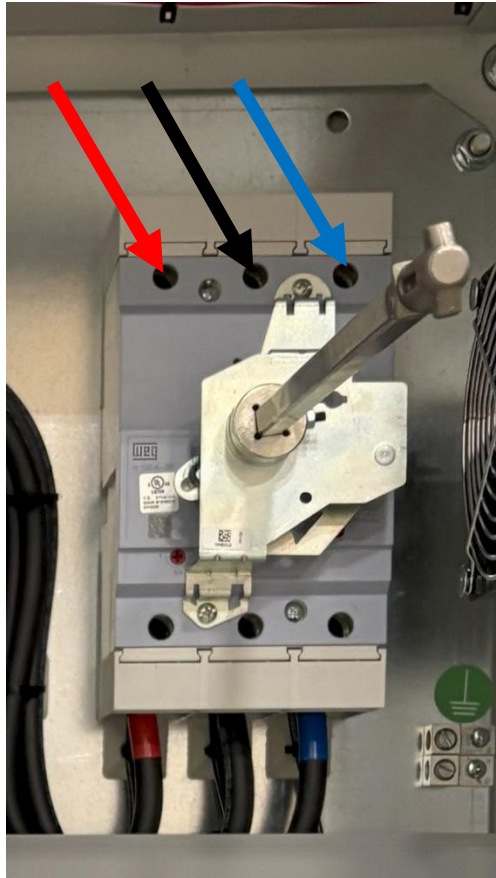


Figure 2 Main Breaker

Control Wiring

For the external shutdown functions listed below, complete the control wiring as shown in the associated diagrams and notes.

1. Jumper Removal

Remove only the jumpers required for the shutdown functions being used in the application.

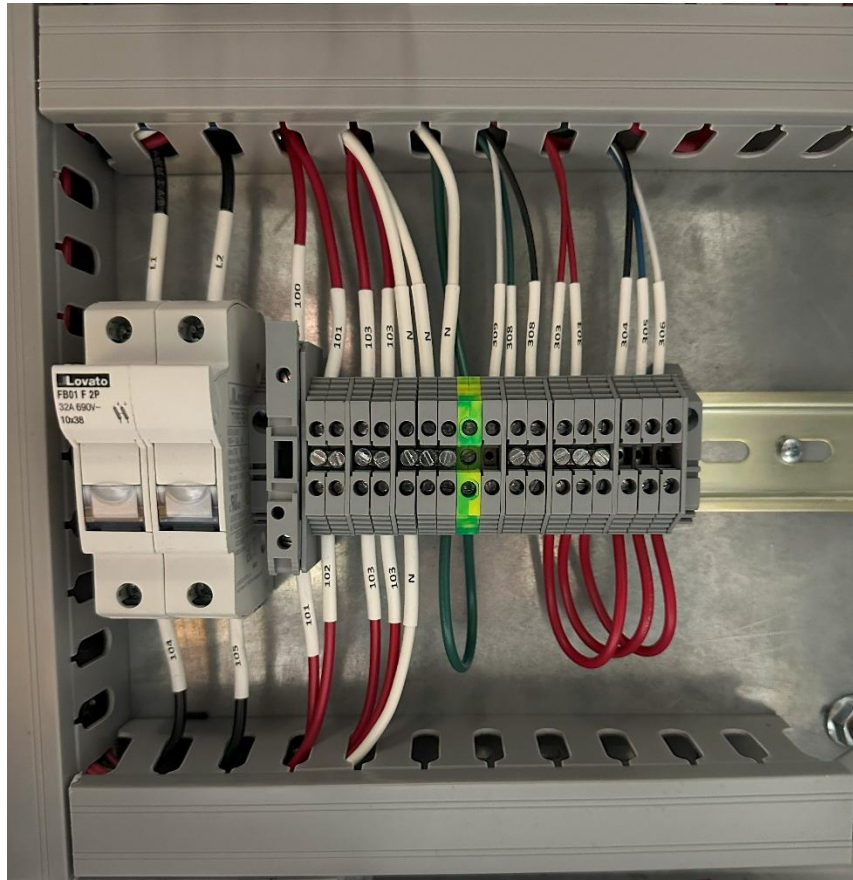


Figure 3 Terminal strip layout

2. Alarm Indications

- The following alarm messages appear on the HMI when the corresponding shutdown condition is active:
 - A750: Underload Shutdown
 - A751: Fault Aux 1 Shutdown
 - A752: Fault Aux 2 Shutdown
 - A753: Alarm Aux 1 Shutdown
 - A754: Alarm Aux 2 Shutdown
 - A755: Overload Shutdown
 - A756: Backspin Timer On
 - A757: Motor Temp Shutdown
 - A758: Vibration X Shutdown
 - A758: Vibration Y Shutdown
 - A759: Intake Pressure Shutdown
 - A760: Both PIDs Enabled (Future)

3. Additional Information

- Refer to the wiring diagrams below for additional installation details and guidance.



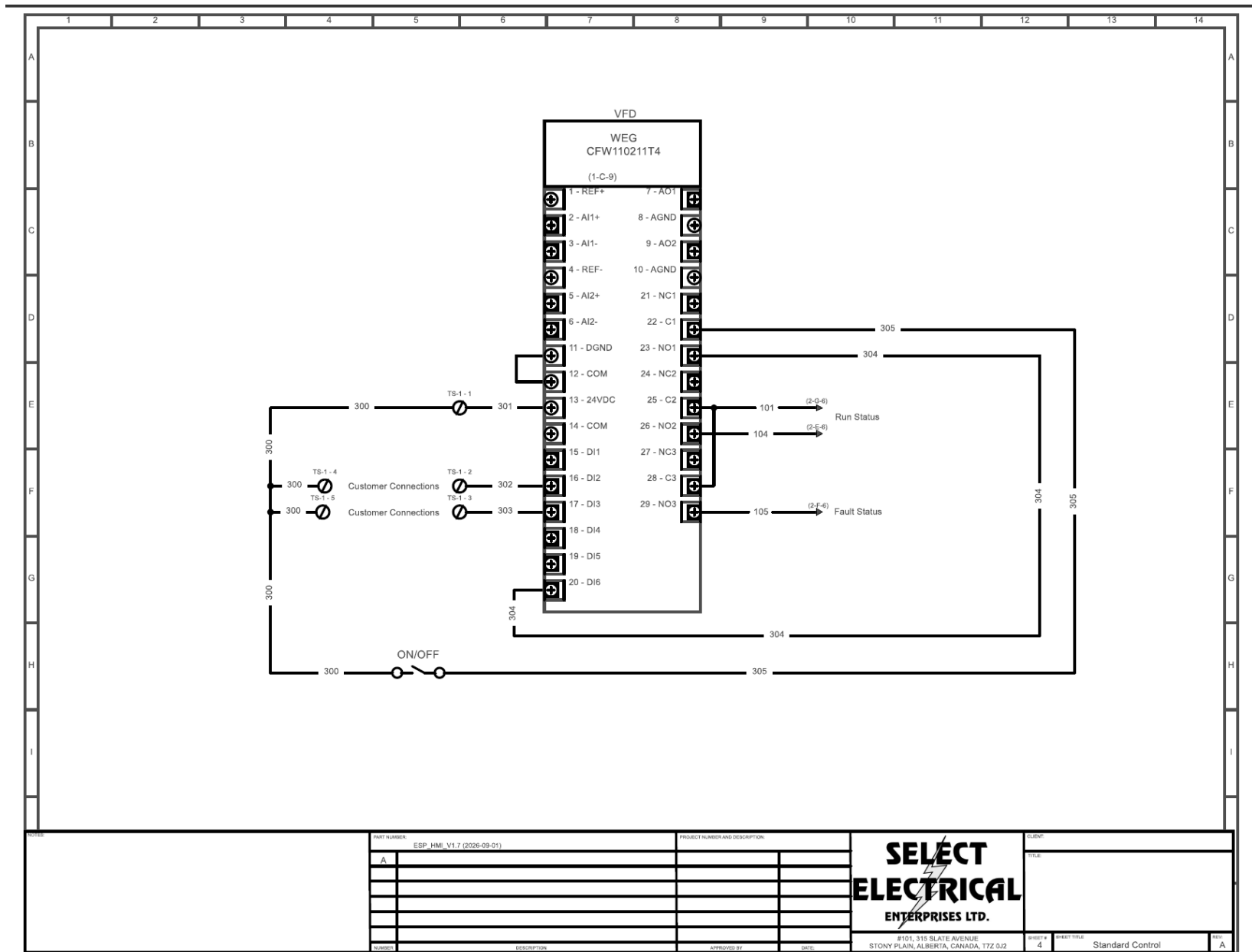


Figure 5b Terminal strip layout

Initial Power Up

During initial startup, the field technician must configure the SE200 using the Field Technician Parameters identified in the startup guide. Parameters highlighted in blue, yellow, and gold are intended for field configuration. Parameters shown in white are factory-set by SEEL and should be changed only after consulting SEEL technical support. A full copy of the startup guide is included with each SE200 unit. Follow the steps below to begin programming the VFD. **Parameter steps #1-#28 are set in SEEL factory and do not need to be set in the field unless otherwise told by SEEL Representative, or in other parts of this manual.**

Steps to access and configure parameters:

1. Access the Parameter Section

- Press **Menu > Oriented Startup**.
- Use the **Up** and **Down** arrows to scroll to **Startup**.
- Set **P0317** to 1 to enter **Oriented Startup**.

2. Complete the Oriented Startup Menu

- Beginning at Step 29, the SE200 VFD guides you through the Oriented Startup sequence (Steps 29-38), shown in **yellow**. When entering motor voltage, frequency, and RPM values, make sure they match the intended maximum operating speed. For example, if the target maximum speed is 65 Hz with a 480 V motor voltage target, use the following settings:
- **P0400** (Motor Rated Voltage) needs to be set for **480v**.
- **P0403** (Motor Rated Frequency) needs to be set for **65hz**.
- **P0402** (Motor Rated RPM) needs to be set at $1800\text{rpm} \times (65\text{hz}/60\text{hz}) = 1950\text{rpm}$.
- Configuring the VFD this way maintains the correct **volts-per-hertz (V/Hz) ratio** across the operating range and helps ensure the drive outputs the correct voltage at the selected speed. Incorrect motor rating values can result in poor flux control, reduced torque, or improper voltage levels at higher frequencies.

3. Continue Programming

- Return to **Menu > All Parameters** to continue the programming procedure using Steps 39-45, shown in **blue**.
- Steps 46-70, shown in **gold**, are configured through the large Weintek HMI on the front of the package. See page 13 of the startup guide for an overview of the panel.

PARAMETERS TO BE SET BY FIELD PERSONNEL (Steps #1-42)					
Step #	Parameter #	Parameter Description	Unit	Default Value	Set Value
1	P0000	Access to Parameters	0-9999	0000	0005
2	P0100	Acceleration Time	Sec	20.0	30.0
3	P0101	Deceleration Time	Sec	20.0	30.0
4	P0132	Max Over Speed Level	0-100%	10%	20%
5	P0217	Zero Speed Disable	Disable Magnetization Current	0	1
6	P0220	LOC/REM Selection Source	Always REM	2	1
7	P0221	LOC Reference Select	AI1	0	1
8	P0222	Remote Reference Select	Soft PLC	1	12
9	P0223	LOC FWD/REV	Always FWD	2	0
10	P0224	LOC Run/Stop Selection	SoftPLC	0	5
11	P0225	LOC Jog Selection	Disable	1	0
12	P0226	REM FWD/REV	Always FWD	4	0
13	P0227	REM Run/Stop Selection	SoftPLC	1	5
14	P0228	REM Jog Selection	Disable	2	0
15	P0229	Stop Mode Selection	Coast to Stop	0	1
16	P0233	AI1 Signal Type	0-10vdc (0) or 4-20mA (1)	0	0
17	P0238	AI2 Signal Type	0-10vdc (0) or 4-20mA (1)	0	Either 0 or 1
18	P0263	DI1 Function	NOT USED	1	0
19	P0264	DI2 Function	PLC Use (AUX 1 SD)	8	21
20	P0265	DI3 Function	PLC Use (AUX 2 SD)	0	21
21	P0266	DI4 Function	NOT USED	0	0
22	P0267	DI5 Function	NOT USED	10	0
23	P0268	DI6 Function	General Enable	14	2
24	P0275	DO1 Function (RL1)	Soft PLC (General Enable)	13	28
25	P0276	DO2 Function (RL2)	No Fault/Alarm	2	36
26	P0277	DO3 Function (RL3)	Run	1	11
27	P0320	Fly Start/Ride-Through	Fly Start/Ride-Through	0	3
28	P0353	IGBT's / Air Overtemp Config	Under Temp Disable	0	4
29	P0202	Type of Control	V/f 60 Hz	0	0
30	P0296	Line Rated Voltage	200-690V	According to inverter model	Set to line voltage

31	P0298	Application	Normal Duty or Heavy Duty	0	0
32	P0398	Motor Service Factor	1.00 to 1.50	1.00	As nameplate
33	P0400	Motor Rated Voltage	0 to 690V	According to inverter model	As Desired
34	P0401	Motor Rated Current	0 to 1.3 x I nom	1.0	As nameplate
35	P0401	Motor Rated Speed	RPM	1750	As Desired
36	P0403	Motor Rated Frequency	0 to 300Hz	60	As Desired
37	P0404	Motor Rated Power	0 to 3400HP, 2500KW	According to inverter model	As nameplate
38	P0407	Motor Rated Power Factor	0.5 to 0.99	0.68	As nameplate
39	P0135	Maximum Output Current	To 2 x I nom-HD	1.5 x I nom-HD	Set to desired maximum output current
40	P0156	Overload Current 100% Speed	0.1 to 1.5 x Inom-ND	1.05 x Inom-ND	As per motor service factor, adjust if required (SF=1.0 – 1.15 x Inom-ND) (SF=1.15 – 1.25 x Inom-ND)
41	P0157	Overload Current 50% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.9 x Inom-ND	As per motor service factor, adjust if required
42	P0158	Overload Current 5% Speed	0.1 to 1.5 x Inom-ND DO NOT ADJUST UNLESS REQUIRED	0.65 x Inom-ND	As per motor service factor, adjust if required
43	P0159	Motor Thermal Class	Class 5,10,15,20,25,30,35,40,45	1 – Class 10	3 – Class 20
44	P0340	Auto Reset Time (VFD Faults)	0-3600s	0	Set to desired reset time
45	P0357	Line Phase Loss Time	0-60sec	3 sec	Set to 0 seconds

46	P1001	Soft PLC Command	Ensure this is set to "Run Program"	0	1
47	P1010	Back Spin Timer	0-9999 mins	0	Desired Back Spin Lockout Time
48	P1016	Back Spin Time Elapsed	Read Only Parameter	n/a	n/a
49	P1023	Minimum Frequency	0-400	0	Desired Minimum Frequency
50	P1024	Maximum Frequency	0-400	0	Desired Maximum Frequency
51	P1012	Underload Debounce Setpoint	0-3000	0	Desired Underload Shutdown Debounce Setpoint
52	P1011	Underload Shutdown Setpoint (Amps)	0-3000	0	Desired Underload Shutdown Setpoint
53	P1013	Shutdown Bypass Time	0-9999	0	Desired Auto Restart Wait Time
54	P1015	Shutdown Bypass Time Elapsed	Read Only Parameter	N/A	N/A
55	P1029	Speed Setpoint	0-60Hz	0	Speed setpoint value
56	P1031	Underload Startup Bypass Time	0-3000 sec	0	Set to Desired time required to bypass underload during start up.
57	P1033	Motor Temp Shutdown Enable	0-1	0	Set to 1 to enable shutdown
58	P1034	Motor Temperature	0-3000	N/A	Displays motor Temp in C
59	P1035	Motor Temperature Shutdown Setpoint	0-3000	0	Set to desired temp S/D
60	P1036	Vibration Shutdown Enable	0-1	0	Set to 1 to enable shutdown
61	P1037	Vibration X Shutdown Setpoint	G	0-32767	Set to desired shutdown
62	P1038	Vibration Y Shutdown Setpoint	G	0-32767	Set to desired shutdown
63	P1039	Vibration X	G	0-32767	Displays Vibration X
64	P1040	Vibration Y	G	0-32767	Displays Vibration Y

65	P1041	Intake Pressure Shutdown Enable	0-1	0	Set to 1 to enable shutdown
66	P1042	Intake Pressure Shutdown Setpoint	PSI	0-32767	Set to desired shutdown
67	P1043	Intake Pressure	PSI	0-32767	Displays the Intake Pressure in PSI
68	P1044	Motor Temp S/D Debounce	Sec	0-32767	Set to desired shutdown debounce time
69	P1045	Vibration S/D Debounce	Sec	0-32767	Set to desired shutdown debounce time
70	P1046	Intake Pressure S/D Debounce	Sec	0-32767	Set to desired shutdown debounce time

Table 1 Startup Guide

Main Screen

The Main Screen is the central navigation hub for programming and operating the SE200 ESP. From this screen, users can access the primary setup, status, trending, alarm, and technician functions.

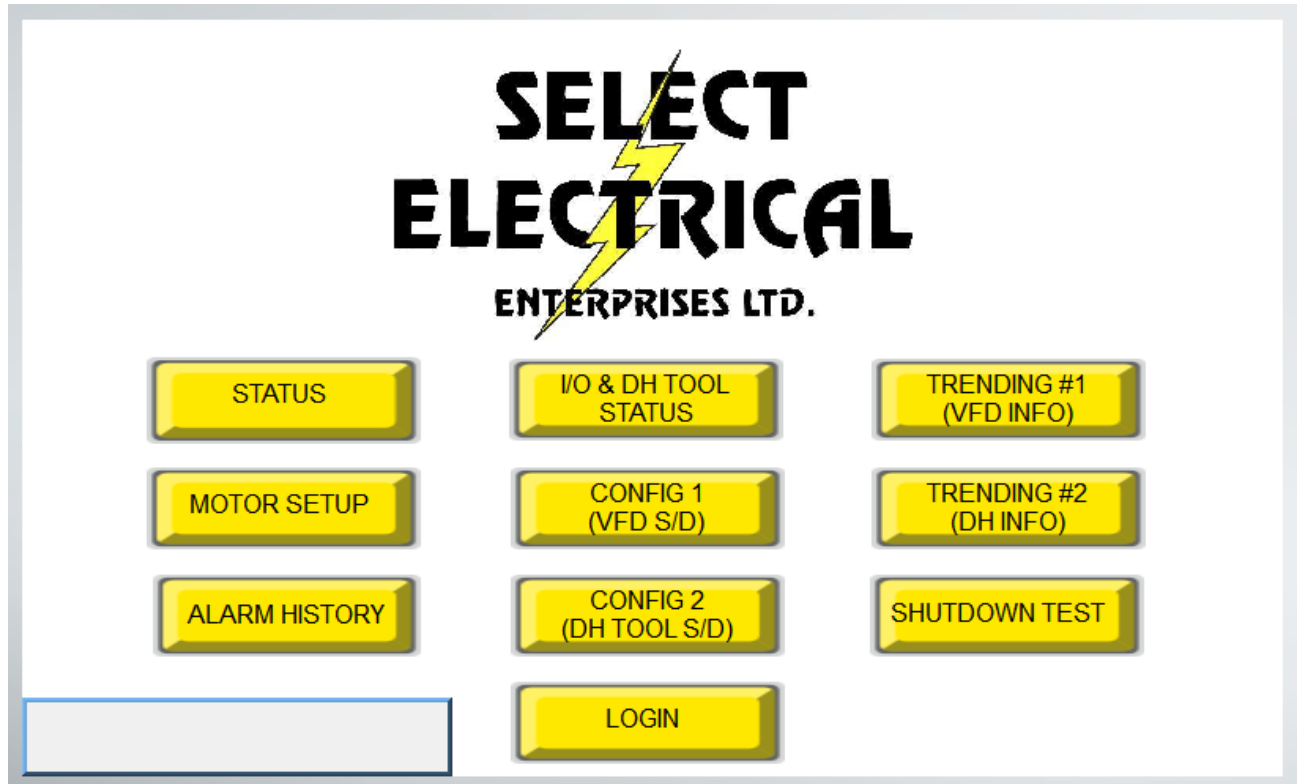


Figure 1 Main Screen

The following list describes the function of each button:

- **STATUS** – Opens the Status page, which serves as the primary operating screen for the SE200 ESP.
- **MOTOR SETUP** – Opens the motor configuration page, where the technician can enter motor data and perform motor auto-tuning.
- **I/O STATUS** – Opens the I/O Status page, which is useful for troubleshooting faults and verifying system inputs and outputs.
- **CONFIG 1 (VFD S/D)** – Opens the VFD shutdown and debounce configuration page, where technicians can set delay timers and startup bypass values.
- **CONFIG 2 (DH TOOL S/D)** – Opens the downhole tool shutdown and control page, where technicians can enable or disable shutdowns, set thresholds, and view live values.
- **SHUTDOWN TEST** – Opens the shutdown bypass page, where auxiliary shutdowns can be temporarily bypassed for testing without shutting down the well.

- **TRENDING #1 (VFD INFO)** – Opens the VFD trending page, which displays operating data over time.
- **TRENDING #2 (DH TOOL INFO)** – Opens the downhole tool trending page, which displays DH tool data over time.
- **ALARM HISTORY** – Opens the Alarm History page.
- **LOGIN** – Opens the login pop-up, where the technician can sign in to access technician-level parameters. See the Login section for the password.

Note: Menus shown in orange are available only after Technician Mode is enabled from the login pop-up menu.

Status Page

The Status Page is the primary operating screen for the SE200 ESP. From this screen, the operator can start or stop the unit, reset active faults or alarms, adjust PID setpoints, view suction or discharge pressure, monitor key motor values, and review any active faults or alarms. The page also provides access to the Main menu and Alarm History.

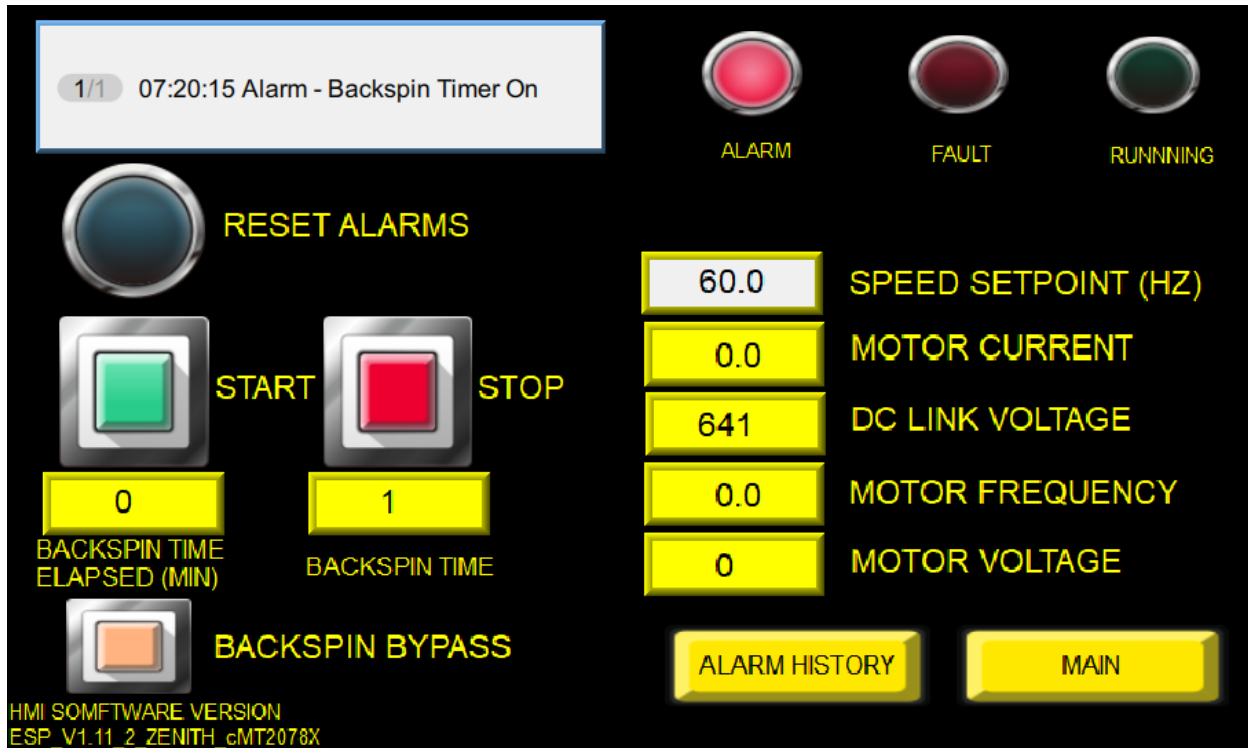


Figure 2 Status page

- **START** – Starts the pumping unit. If the unit is in a faulted state, the fault must be reset before startup is allowed.
- **STOP** – Stops the pumping unit.
- **MAIN** – Opens the Main menu.
- **ALARM HISTORY** – Opens the Alarm History page.
- **SPEED SETPOINT** – Enter the target operating speed in Hz for the ESP.
- **MOTOR CURRENT** – Displays the live motor current value. This value is also shown on Trending Page #1.
- **DC LINK VOLTAGE** – Displays the live DC link voltage.
- **MOTOR FREQUENCY** – Displays the live motor output frequency. This value is also shown on Trending Page #1.
- **MOTOR VOLTAGE** – Displays the live motor output voltage. This value is also shown on Trending Page #1.

Motor Setup

The Motor Setup page provides technicians with a simplified view of the motor parameters required for configuration, all arranged on a single screen.

The screenshot displays the Motor Setup interface with the following elements:

- Motor Service Factor (1.0-1.50):** Input field containing 1.00.
- Motor Rated Efficiency (50.0-99.9%):** Input field containing 67.0.
- Motor Rated Voltage:** Input field containing 460.
- Motor Rated Current:** Input field containing 2.0.
- Motor Rated Hz:** Input field containing 60.
- Acceleration Time (Sec):** Input field containing 20.0.
- Deceleration Time (Sec):** Input field containing 20.0.
- Motor Rated Power:** Dropdown menu showing 1.5HP 1.1KW.
- Line Rated Voltage:** Dropdown menu showing 440-460V.
- Navigation Buttons:** Two yellow buttons labeled MAIN and STATUS.

Figure 3 Motor setup page

- **MOTOR PARAMETERS** – Enter the motor nameplate data in the corresponding fields above.
- **ACCELERATION TIME** – Sets the time required for the ESP to accelerate linearly from 0 to maximum speed.
- **DECELERATION TIME** – Sets the time required for the ESP to decelerate linearly from maximum speed to 0.

I/O & DH Tool Status

The I/O Status page provides a detailed view of SE200 input and output states, along with incoming pressure values displayed as percentages. This page is intended primarily for technician troubleshooting and diagnostics.

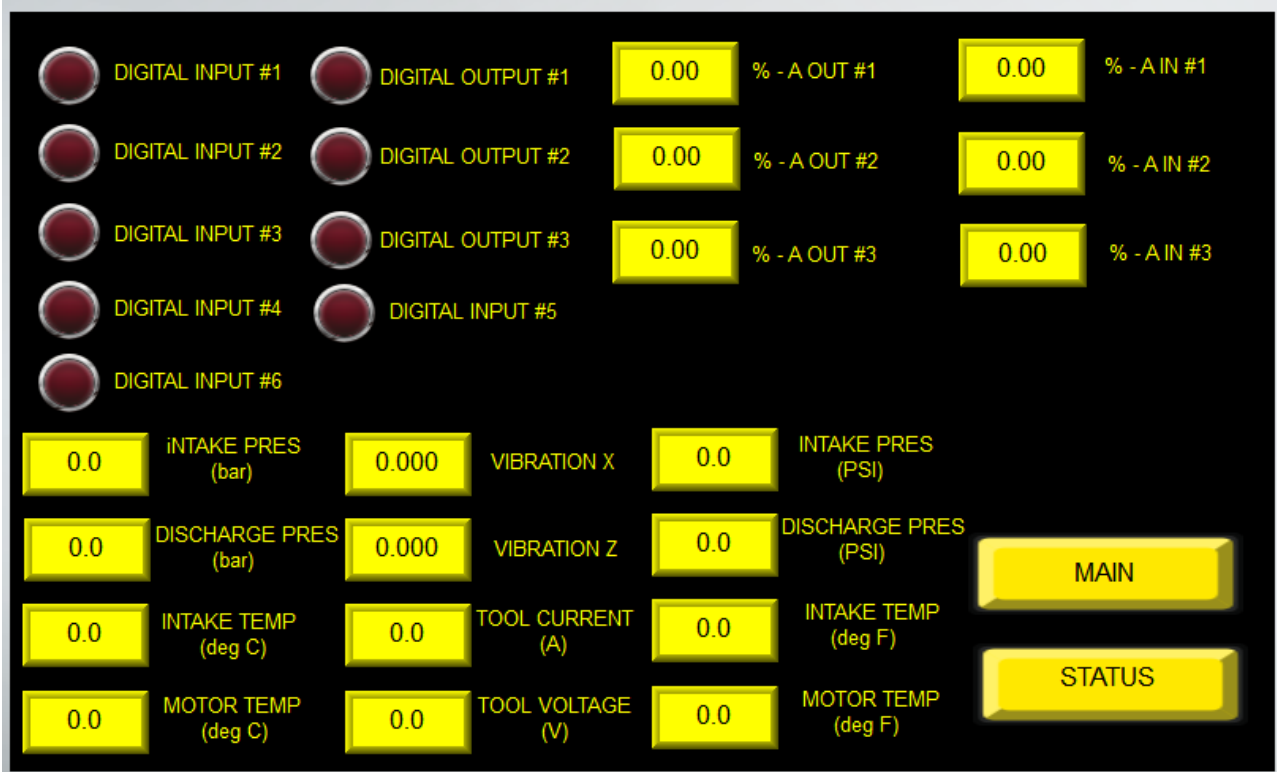


Figure 4 I/O Status page

Config 1 (VFD S/D)

The Config 1 (VFD S/D) page allows technicians to configure VFD shutdown debounce timers and startup delay timers for the application. This page is available only when Technician Mode is enabled.

0.0	Underload Setpoint (A)	3.0	Maximum Current (A)
0	Underload Debounce (Sec)	0.00	Auto Torque Boost (0.0-1.0)
10	Underload Startup Bypass Time (Sec)		
5.0	Overload Setpoint (A)		
0	Overload Debounce (Sec)		
1	Backspin Timer (MIN)		
0.0	Minimum Speed (Hz)		MAIN
60.0	Maximum Speed (Hz)		STATUS

Figure 5 S/D Debounce & Startup page

- **UNDERLOAD SETPOINT** – Sets the shutdown threshold for pump underload, in amps.
- **UNDERLOAD DEBOUNCE** – Sets the delay time before an underload condition triggers a shutdown.
- **UNDERLOAD STARTUP BYPASS TIME** – Sets the startup bypass period for underload protection, allowing the pump time to load before shutdown logic becomes active.
- **OVERLOAD SETPOINT** – Sets the shutdown threshold for pump overload, in amps.
- **OVERLOAD DEBOUNCE** – Sets the delay time before an overload condition triggers a shutdown.
- **BACKSPIN TIMER** – Sets the pump backspin lockout time.
- **MINIMUM SPEED** – Sets the minimum operating speed of the ESP. This is typically the minimum speed required to properly lubricate the system; consult the pump technician for the correct value.
- **MAXIMUM SPEED** – Sets the maximum operating speed of the ESP motor. Consult the pump technician for the correct application-specific value.
- **MAXIMUM CURRENT** – Sets the VFD current limit.
- **AUTO TORQUE BOOST** – Enables or disables automatic torque boost.

Config 2 (DH Tool S/D)

The Config 2 (DH Tool S/D) page allows technicians to configure downhole tool shutdown settings and debounce timers for the application. This page is available only when Technician Mode is enabled.

TO ENABLE SHOT DOWN TURN ON TOGGLE SWITCH

0.0	MOTOR TEMP S/D SETPOINT	MOTOR TEMP S/D ENABLE	MOTOR TEMP S/D DEBOUNCE
0.0	MOTOR TEMP (C)		0
0.000	VIBRATION-Vx S/D SETPOINT	VIBRATION X/Y S/D ENABLE	VIBRATION S/D DEBOUNCE
0.000	VIBRATION-Vy S/D SETPOINT		0
0.000	VIBRATION X (Vx)		INTAKE PRESSURE S/D DEBOUNCE
0.000	VIBRATION Z (Vz)	INTAKE PRESSURE S/D ENABLE	0
0.0	INTAKE PRESSURE S/D SETPOINT		MAIN
0.0	INTAKE PRESSURE (PSI)		STATUS

Figure 6 Suction S/D & Control Menu page

- **MOTOR TEMP S/D SETPOINT** – Sets the motor temperature shutdown threshold.
- **MOTOR TEMP** – Displays the live motor temperature value.
- **VIBRATION Vx S/D SETPOINT** – Sets the high X-axis vibration shutdown threshold.
- **VIBRATION Vz S/D SETPOINT** – Sets the high Z-axis vibration shutdown threshold.
- **VIBRATION X** – Displays the live X-axis vibration value.
- **VIBRATION Z** – Displays the live Z-axis vibration value.
- **VIBRATION X** – Displays the live X-axis vibration value.
- **INTAKE PRESSURE S/D SETPOINT** – Sets the intake pressure shutdown threshold.
- **INTAKE PRESSURE** – Displays the live intake pressure value.
- **MOTOR TEMP S/D ENABLE** – Enables or disables motor temperature shutdown protection.
- **VIBRATION Vx/Vz S/D ENABLE** – Enables or disables vibration shutdown protection.
- **INTAKE PRESSURE S/D ENABLE** – Enables or disables intake pressure shutdown protection.
- **MOTOR TEMP S/D SETPOINT** – Sets the debounce delay for motor temperature shutdown.
- **VIBRATION Vx/Vz S/D DEBOUNCE** – Sets the debounce delay for vibration shutdown protection.
- **INTAKE PRESSURE S/D DEBOUNCE** – Sets the debounce delay for intake pressure shutdown protection.

Shutdown Test

The Shutdown Test menu allows users to temporarily bypass selected shutdown functions for a defined period. This makes it possible to test shutdown inputs without shutting down the equipment or waiting for the backspin bypass timer to expire.

Figure 7 Discharge S/D & Control page

- **SHUTDOWN BYPASS TIMER** – Sets the auxiliary shutdown bypass duration used during testing.
- **TIME ELAPSED** – Displays the elapsed time since the shutdown bypass was started.
- **START SHUTDOWN BYPASS BUTTON** – Starts the selected shutdown bypass.
- **TOGGLE SWITCH** – Selects which auxiliary shutdown is bypassed. Right selects Auxiliary Shutdown 1, and left selects Auxiliary Shutdown 2.

Trending #1 (VFD Info)

The Trending #1 page displays motor voltage, frequency, and current over time. The trend can be scrolled to review earlier data, and the data can also be downloaded to a PC using the USB port on the back of the unit.

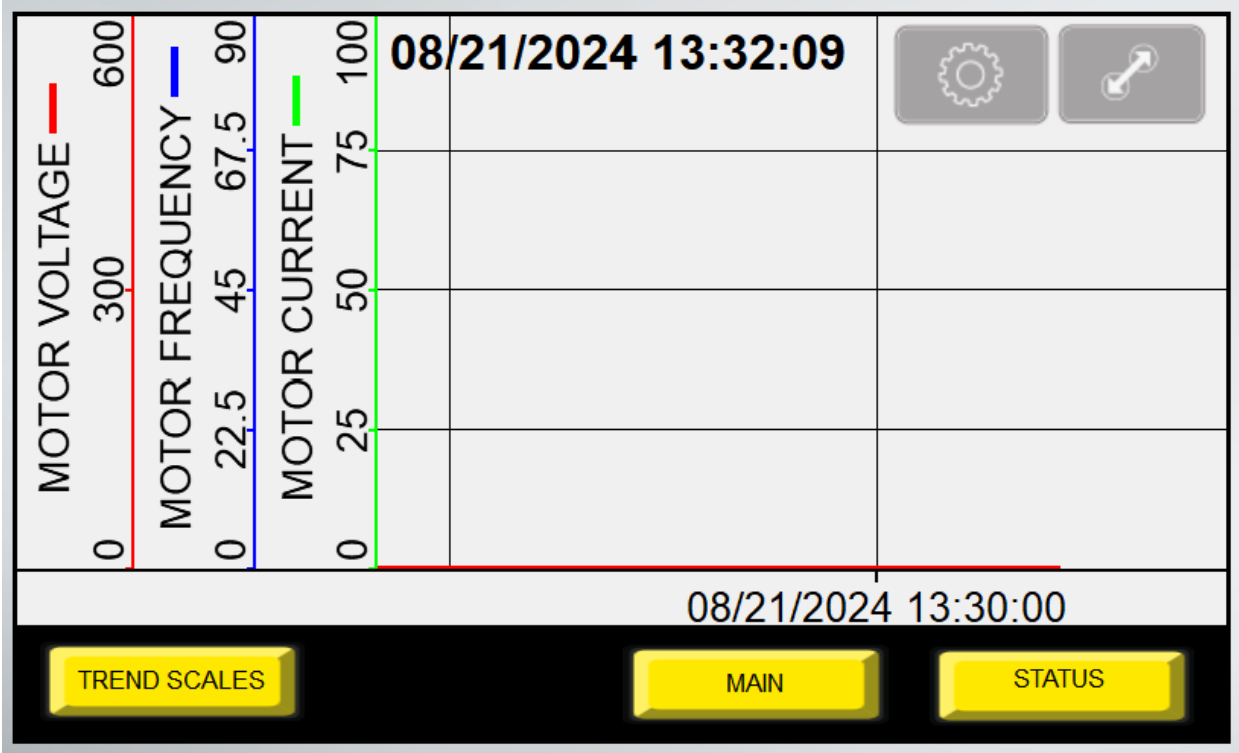


Figure 8 Trending #1 page

Trending #2 (DH Tool Info)

The Trending #2 page displays intake pressure, vibration Z, vibration X, and motor temperature over time. The trend can be scrolled to review earlier data, and the data can also be downloaded to a PC using the USB port on the back of the unit.

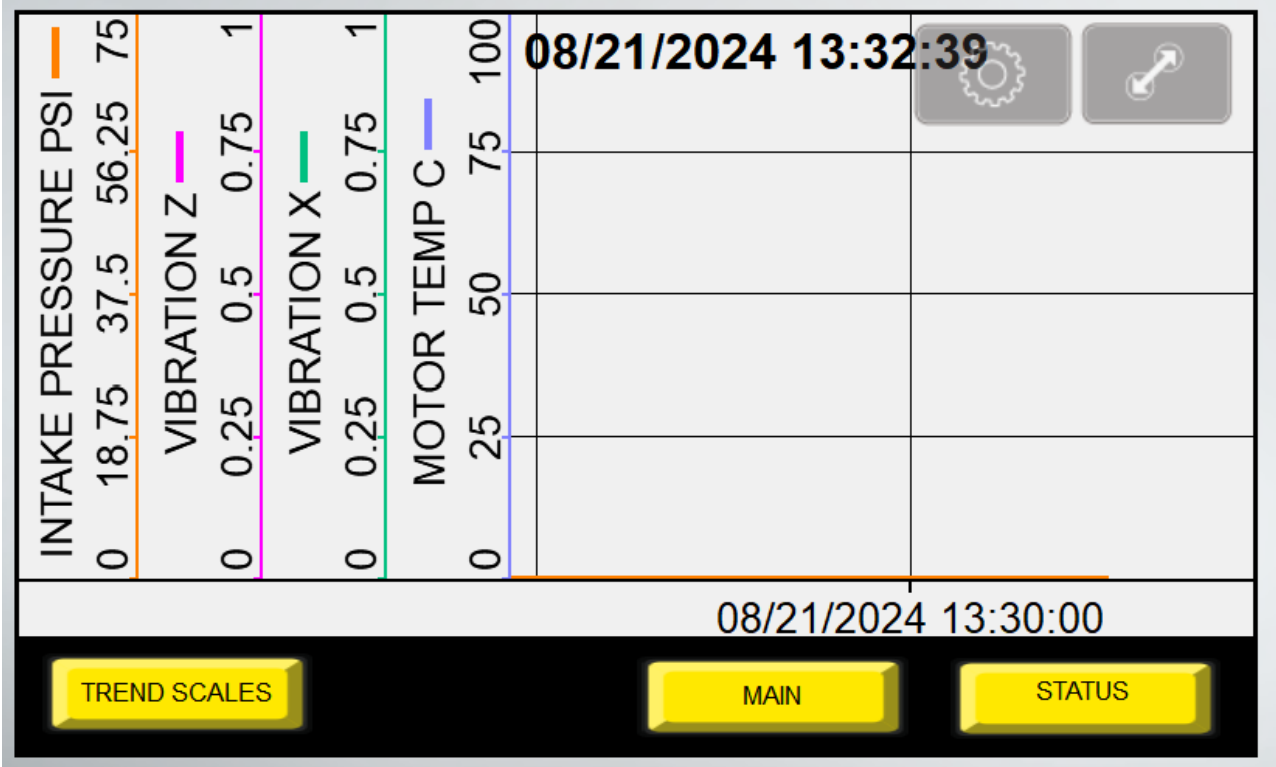


Figure 9 Trending #2 page

Trending Scales

The Trending Scales page allows the user to adjust scaling values based on the ranges of the suction and discharge pressure transmitters and the motor current. **Note: This page is accessible only from the Trending #1 and Trending #2 menus.**

The screenshot shows a black background with yellow text and buttons. On the left, there are eight input fields with yellow borders, each containing a numerical value. To the right of each input field is a label in yellow text. On the far right, there are four yellow buttons stacked vertically.

Input Value	Label
0	MOTOR CURRENT MIN SCALE
100	MOTOR CURRENT MAX SCALE
0	VIBRATION X Vx MIN SCALE
1	VIBRATION X Vx MAX SCALE
0	VIBRATION Z Vz MIN SCALE
1	VIBRATION Z Vz MAX SCALE
0	INTAKE PRESSURE MIN SCALE
75	INTAKE PRESSURE MAX SCALE

Navigation buttons on the right:

- TRENDING #1 (VFD INFO)
- TRENDING #2 (DH TOOL INFO)
- MAIN
- STATUS

Figure 10 Trending Scales page

- **Alarm History Display** – Displays the date and time associated with each alarm event.
- **RESET FAULT/ALARM** – Clears any active fault or alarm on the ESP.
-

Alarm History

The Alarm History page provides a record of ESP faults and alarms. Alarm history can also be exported to a USB device for download and review.

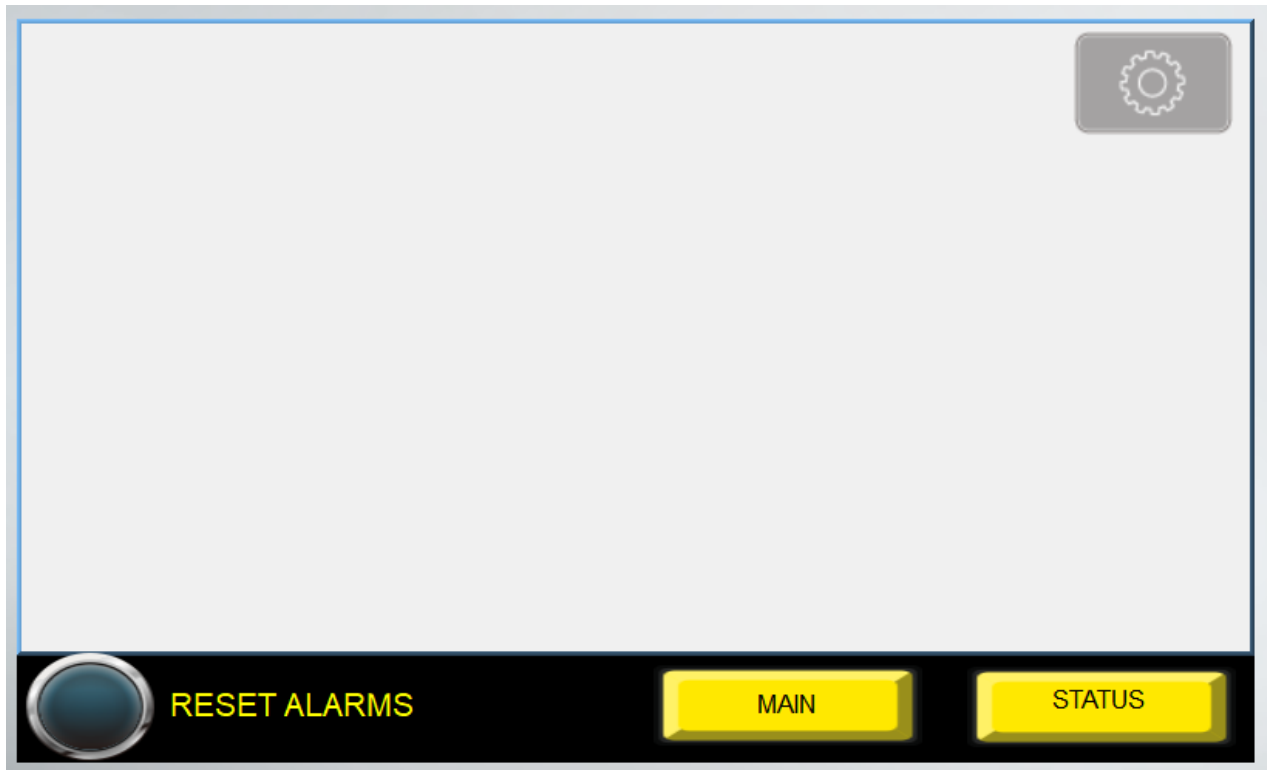


Figure 11 Alarm History page

- **Alarm History Display** – Displays the date and time associated with each alarm event.
- **RESET FAULT/ALARM** – Clears any active fault or alarm on the ESP.

Login Pop-up

Accessible from the Main Menu, the Login pop-up allows the technician to sign in and enable technician-level parameters. **The technician password is 4444.** Enter the password and press Enter on the keypad. After the keypad closes, press the Login button. If the status below the password field indicates success, Technician Mode has been enabled. Press the **red X** to close the pop-up and continue with parameter entry.



Figure 12 Login page

Troubleshooting

For common faults/alarms and their possible causes see table starting on page 13.

Faults/Alarms	Description	Possible Causes
F006 Imbalance or Input Phase Loss	Mains voltage imbalance too high or phase missing in the input power supply. Note: - If the motor is unloaded or operating with reduced load this fault may not occur. - Fault delay is set at parameter P0357. P0357 = 0 disables the fault. - If single-phase power supply is used, this fault must be disabled.	Phase missing at the inverter's input power supply. Input voltage imbalance >5 %. For the Frame Size E: Phase loss at L3/R or L3/S may cause F021 or F185. Phase loss at L3/T will cause F006. For frame sizes F and G: Pre-charge circuit fault.
F021 DC Link Undervoltage	DC Link undervoltage condition occurred	The input voltage is too low, and the DC bus voltage dropped below the minimum permitted value (monitor the value at parameter P0004): DC BUS Voltage < 446 V - For a 440-460 V input voltage (P0296 = 3). DC BUS Voltage < 487 V - For a 480 V input voltage (P0296 = 4). Phase loss in the input power supply. Pre-charge circuit failure. Parameter P0296 was set to a value above of the power supply rated voltage. When free spooling can increase P0169 and P0170 as needed depending on speed.

F022 DC Link Overvoltage	DC link overvoltage condition occurred.	The input voltage is too high, and the DC bus voltage surpassed the maximum permitted value: DC Bus Voltage > 400 V - For 220-230 V input models (P0296 = 0). DC Bus Voltage > 800 V - For 380-480 V input Inertia of the driven load is too high, or deceleration time is too short. Wrong settings for parameters P0151, or P0153, or P0185. Ensure P0185 and P0186 are set to 770 and 40 respectively. When free spooling can increase P0169 and P0170 as needed depending on speed.
F048 IGBT Overload Fault	An IGBTs overload fault occurred.	Inverter output current is too high.
F070 Overcurrent / Short-circuit	Overcurrent or short-circuit detected at the output, in the DC bus, or at the braking resistor.	Short-circuit between two motor phases. Short-circuit between the connection cables of the dynamic braking resistor. IGBT modules are shorted. Disconnect Motor try running VFD if issue persists replace inverter, if problem goes away check motor and cable for shorts perform a megger test.
F071 Output Overcurrent	The inverter output current was too high for too long.	Excessive load inertia or acceleration time too short. Settings of P0135 or P0169 and P0170 are too high
F072 Motor Overload	The motor overload protection operated. Note: It may be disabled by setting P0348 = 0 or 3	Settings of P0156, P0157 and P0158 are too low for the used motor. Motor shaft load is excessive.

F0150 Motor Overspeed	Overspeed fault. It is activated when the real speed exceeds the value of $P0134 \times (100 \% + P0132)$ for more than 20 ms.	Wrong settings of P0161 and/or P0162. Problem with the hoist-type load. Adjust P0132 higher or increase P01011.
A163/164 Break Detect AI1/AI2	It indicates that the AI current (4-20 mA or 20-4 mA) reference is out of the 4 to 20 mA range.	Broken AI cable. Bad contact at the signal connection to the terminal strip. Blown control fuse on terminal strip for AI circuit. If there is no AI going to that input set the corresponding input P0233-AI1 or P0238-AI2 to 0.
A181 Invalid Clock Value	Invalid clock value alarm	It is necessary to set date and time at parameters P0194 to P0199. (Try this first if not then check battery) Keypad battery is discharged, defective, or not installed.
F185 Pre-charge Contactor Fault	It indicates fault at the pre-charge contactor	Pre-charge contactor defect. Open command fuse. Phase loss in the input L1/R or L2/S. P0355 = 1 (Incorrect setting for model's frame E supplied by the DC link. For those models, the setting should be P0355 = 0.

For CFW11 Alarm & Fault List refer to page 6-2 of the CFW-11 User Manual Provided

Alarms

A750 – Underload Shutdown
A751 – Fault Aux 1 Shutdown
A752 – Fault Aux 2 Shutdown
A753 – Alarm Aux 1 Shutdown
A754 – Alarm Aux 2 Shutdown
A755 – Overload Shutdown
A756 – Backspin Timer On
A757 – Motor Temp Shutdown

A758 – Vibration X Shutdown
A758 – Vibration Y Shutdown
A759 – Intake Pressure S/D
F760 – Both PIDs Enabled

Digital Inputs and Outputs

DI1 – Not Used
DI2 – Fault Aux 1 Shutdown
DI3 – Fault Aux 2 Shutdown
DI4 – Not Used
DI5 – Not Used
DI6 - General Enable (Normally Open)

DO1 – Soft PLC (N/O)
DO2 – Run (N/O)
DO3 – No Fault/Alarm (N/O)

Analog Input / Output Configuration (Dipswitch's Located at top of control board)

S1.1 (Analog Output 1): Dip Switch Off - 4-20mA, Dip Switch On: 0-10vdc
S1.2 (Analog Output 2): Dip Switch Off - 4-20A, Dip Switch On: 0-10vdc
S1.3 (Analog Input 2): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA
S1.4 (Analog Input 1): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

Technical Support

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